OOS Number: OOS- 262289

Phase I

Section A – Test Details					
Initiator:	Varsha Subramanian (Written by: Qiyue Chen)	**Test Date:**	01-Oct-2026	**Date Initiated:**	01-Oct-2026
Date of Incident:	01-Oct-2026	**Test Name:**	Scan RDI Sterility Test	**Sample ID:**	ETX-260929-0931
Name of Analyst who Performed the Test:	Prepping Analyst: Sasha Allen (SAA) Processing Analyst: Ellen Bennett (ELB) Changeover Analyst: Karla Silva (KSM) Reading Analyst: Varsha Subramanian (VV)	**Sample / Active Name**	Tirzepatide/ Levocarnitine 17mg/ 100mg/mL (1ml)	**Lot #:**	20260928@54
		Dosage Form	Injectable		
Description of Incident:	On 01Oct26, sample ETX-260929-0931 was found positive for viable microorganisms after Scan RDI sterility testing.	☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method # MICRO-SOP-12 (16) ENG-SOP-4 (05)	Effective Date: 24Jul26 24Jul26	SOP / Test Method Rev: Rev: 16 Rev: 05	Limits/Specification: <1 event	
Manager notified	Manager/designee name: Kathan Parikh	Date: 01-Oct-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑Yes ☐ No ☐ N/A	Yes, analysts Sasha Allen, Ellen Bennett, Karla Silva, and Varsha Subramanian were interviewed comprehensively.	
Correct samples analyzed?	☑Yes ☐ No ☐ N/A	Yes, ETX-260929-0931	
Correct SOP / Test Method used/followed?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-12, ENG-SOP-4	
Correct sample collection/technique?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-12, ENG-SOP-4	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑Yes ☐ No ☐ N/A	Yes, See 100126-1040-2 for more information.	
Is the analyst qualified to perform the test?	☑Yes ☐ No ☐ N/A	Yes, all analysts are trained and qualified by quality to perform the test.	
Is formula worksheet provided and verified?	☐Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑Yes ☐ No ☐ N/A	Yes, Information is available in Eagle Trax Sample Location History under ETX-260929-0931	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐Yes ☑ No ☐N/A		
Was the sample transported appropriately?	☑Yes ☐ No ☐N/A		
Are the other test results from this run under investigation?	☐Yes ☑ No ☐N/A		

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Appropriate/correct glassware used?	☐Yes ☐No ☑N/A		
Are reagents stored in proper condition and not expired?	☑Yes ☐No ☐N/A	Scan Consumables: See the attached data packet Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs	Scan Consumables: See the attached data packet Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs
Standard Used	P. aeruginosa	05162026-9027-PA	15May28
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐Yes ☐No ☑N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐Yes ☐No ☑N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑Yes ☐No ☐N/A	E001040	Oct 2026
Is the clean room facility certified?	☑Yes ☐No ☐N/A	E001978 (CR144) E001979 (CR145)	Nov 2026 Nov 2026
System Performance Check (if applicable)?	☑Yes ☐No ☐N/A	E001040	Oct 2026
System Suitability RSD < 2% (PDA)?	☐Yes ☐No ☑N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐Yes ☐No ☑N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☑Yes ☐No ☐N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑Yes ☐No ☐N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑Yes ☐No ☐N/A	Incubator E001034 (Sensor E001501) Incubator E001031 (Sensor E001505) BSC E001938 BSC E001937	Aug 2027 Feb 2027 Aug 2027 Feb 2027 Jul 2027 Jul 2027

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Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not Applicable	Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☐ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A QYC 21Oct26

Section D – Phase I Summary

Phase I Summary:

All analysts involved in the preparation, processing, changeover, and reading of the sample – Sasha Allen, Ellen Bennett, Karla Silva, and Varsha Subramanian – were interviewed comprehensively. Their responses are documented throughout this report. This investigation was performed in accordance with MICRO-SOP-53, Sterility Test Out-of-Specification Investigation Procedure.

Upon arrival, the sample was stored refrigerated in accordance with the Client's instructions. Prepping analyst - Sasha Allen - and processing analyst - Ellen Bennett - verified the physical integrity and packaging of the sample throughout both the preparation and processing stages. No leaks, cracks, or visible turbidity were observed prior to testing, verifying container integrity.

All reagents and supplies mentioned in the material section above were stored according to suppliers' recommendations, and their integrity was visually verified before utilization. Moreover, each reagent, media, and consumable possessed valid expiration dates. The functionality of all equipment was confirmed by reviewing continuous environmental and operational monitoring data.

During the preparation phase, Sasha Allen disinfected the sample containers using acidified bleach and placed them into a pre-disinfected storage bin. On 01Oct26, prior to sample processing, Ellen Bennett performed a second disinfection with acidified bleach, allowing a minimum contact time of 10 minutes before transferring the samples into the cleanroom suites. A final disinfection step was completed immediately before introducing the sample into the ISO 5 Biological Safety Cabinet (BSC E001938) located in the ISO 7 cleanroom (145). All activities were performed in accordance with MICRO-SOP-12, Rapid Scan RDI® Test using FIFU Method.

Cleanroom L-Suite, used for both sample processing and changeover, consists of ISO 7 cleanroom 145, ISO 7 buffer room 144, ISO 8 anteroom 143, and outermost ISO 8 room 142. Outward-cascading positive air pressure differentials are continuously maintained across the suite to ensure controlled unidirectional airflow.

The ISO 5 BSC E001938 located in Cleanroom 145 (CR145) and the ISO 5 BSC E001937 located in Cleanroom 144 (CR144) were both thoroughly cleaned and disinfected prior to their respective procedures in accordance with MICRO-SOP-9, Cleaning and Disinfecting Procedure for Microbiology. Both BSCs were certified and approved by the Engineering and Quality Assurance teams prior to testing.

Pre-labeling and filtration activities were performed by Ellen Bennett on 01Oct26 within the ISO 5 biosafety cabinet (BSC E001938) located in the ISO 7 cleanroom (145). The changeover procedure was subsequently performed by analyst Karla Silva on the same date within the ISO 5 biosafety cabinet (BSC E001937) located in Cleanroom 144 of L-Suite. The reading analyst, Varsha Subramanian, confirmed that the ScanRDI instrument E001040 was set up as per ENG-SOP-4 (Scan RDI® System – Operations (Standard C3 Quality Check and Microscope Setup and Maintenance)), and the negative control and positive control for analyst Ellen Bennett yielded expected results.

On 01Oct26, a rapid sterility test was performed on the sample using the ScanRDI method under test record 100126-1040-2. The sample was prepared by analyst Sasha Allen, processed by Ellen Bennett (39th sample processed), changeover performed by Karla Silva, and subsequently read by Varsha Subramanian. The test revealed 15 total events and 4 confirmed microbial events exhibiting Cocci shaped morphology, see Table 1.

Table 2 presents the environmental monitoring results associated with testing performed on 01Oct26. The environmental monitoring (EM) plates were incubated for no less than 48 hours at 30–35°C and for no less than an additional five days at 20–25°C, as per MICRO-SOP-2, Environmental Monitoring of the Clean-room Facility.

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Phase I Summary (continued):

Environmental Monitoring (EM) Recoveries Review:

ISO 5 BSC and Personnel Evaluation:
Upon review of the environmental monitoring data, absence of microbial recovery (No Growth) was demonstrated across all critical ISO 5 zones and personnel monitoring on the date of testing. Daily personnel monitoring for processing analyst Ellen Bennett and changeover analyst Karla Silva yielded no microbial growth on left and right fingertip touch plates. Daily surface sampling (4 locations) and settling plates within processing BSC E001938 and changeover BSC E001937 yielded no microbial growth.

Cleanroom EM Evaluation:
Weekly surface monitoring of Cleanroom L-Suite conducted on 01Oct26 demonstrated no microbial recovery across all sampled cleanroom surfaces, confirming effective sanitization and surface control.

Weekly active air monitoring of Cleanroom L-Suite conducted on 01Oct26 demonstrated no microbial recovery in ISO 7 cleanroom 145 and ISO 8 anteroom 143; however, 1 CFU under test sample ETX-261012-0430 was recovered from ISO 7 cleanroom 144 Section II (identified as Staphylococcus epidermidis), and 5 CFUs under test sample ETX-261012-0433 were recovered from outermost ISO 8 room 142 (identified as Staphylococcus epidermidis, Pseudomonas parafulva, Bacillus safensis, Corynebacterium xerosis, and Corynebacterium ureicelerivorans).

It is important to emphasize that all sample processing activities were performed strictly within the validated ISO 5 BSC E001938 and changeover activities within ISO 5 BSC E001937, located in the ISO 7 cleanroom (145) and in the ISO 7 cleanroom (144) respectively. There is minimal exposure of the test sample to ambient ISO 8 air, as samples and supplies are transferred in disinfected, closed containers on carts through the layered cleanroom suites. Furthermore, the recoveries occurred in the general cleanroom and lower-classified ISO 8 outer room environments, which are physically segregated from the ISO 5 processing zone by closed doors and an outward-cascading positive air pressure gradient. In conjunction with the complete absence of microbial growth on all analyst glove touch plates, settling plates, and ISO 5 BSC work surfaces throughout testing, there is no plausible contamination transfer pathway from the ambient environment to the test sample.

Monthly cleaning and disinfection of the cleanrooms (ISO 7) and their containing biosafety cabinets (ISO 5) were performed on 27Sep26 for L-Suite, as per MICRO-SOP-9, Cleaning and Disinfecting Procedure for Microbiology. It was documented that all chemical indicators passed,which confirmed the efficient monthly cleaning of all cleanrooms in L-suite.

Analyzing a 6-month sample history for Link Pharmacy (E19112), this specific analyte "Tirzepatide/Levocarnitine 17mg/100mg/mL (1ml)" has had no prior Out-of-Specification (OOS) occurrences under ScanRDI sterility testing.

All samples processed on the same day were also evaluated for evidence of sample-to-sample cross-contamination. The test sample ETX-260929-0931 was the 39th sample processed by Ellen Bennett during the 2nd Shift on 01Oct26 within BSC E001938 (Cleanroom 144, L-Suite). All other samples processed by Ellen Bennett during this shift yielded negative results. One other sample (ETX-260930-0635, OOS-262285) yielded a positive result on 01Oct26. However, that sample was processed by a different processing analyst during a different shift. The two testing events were segregated by shift, processing analyst and primary containment equipment. Furthermore, analyst Ellen Bennett verified that gloves were disinfected between samples and standard aseptic techniques were strictly maintained. Consequently, there is no evidence of sample-to-sample cross-contamination or a shared laboratory vector.

Based on the laboratory investigation, no assignable laboratory cause was identified involving the analyst, instrument, reagents, supplies, or monitored laboratory environment. Therefore, the likelihood of laboratory error contributing to this failure is considered minimal, and the original result is deemed valid.

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Phase I Summary (continued):

N/A QYC 21Oct26

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Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes

☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (*Print*)**:**	Signature/Date:
Qiyue Chen	Qiyue Chen Digitally signed by Qiyue Chen DN: OU=Eagle Analytical Services, O=Strerie Lab, CN=Qiyue Chen, E=qchen@eagleanalytical.com Location: Houston, TX Date: 2026.10.27 11:15:04-05'00' Foxit PDF Editor Version: 13.2.7
Lab Manager/Designee (*Print*)**:**	Signature/Date:
Robin Seymour	

Section E – Quality Disposition

Can investigation be closed?

☐ Yes

☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date:

Table 1: Information for ETX-260929-0931 under investigation

Processing Analyst	Reading Analyst	Sample ID	Events	Confirmed Microbial Events	Morphology Description
Ellen Bennett	Varsha Subramanian	ETX-260929-0931	19	4	***Cocci shaped morphology***

Table 2: Environmental Monitoring from Testing Performed on 01Oct26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst	Day /Week(s)	Observation	EM Plate ETX Number	Microbial ID	Notes
Personnel EM								
Personal (Left Touch and Right Touch)	Daily	01Oct26	ELB	Date of Testing (Scan)	No Growth	Not Applicable	Not Applicable	None
Personal (Left Touch and Right Touch)	Daily	01Oct26	KSM	Date of Testing (Scan C/O)	No Growth	Not Applicable	Not Applicable	None
Biological Safety Cabinet EM for E001938 and E001937								
Surface Sampling of ISO 5 BSC E001938 (4 Locations)	Daily	01Oct26	ELB	Date of Testing (Scan)	No Growth	Not Applicable	Not Applicable	None
Surface Sampling of ISO 5 BSC E001937 (4 Locations)	Daily	01Oct26	KSM	Date of Testing (Scan C/O)	No Growth	Not Applicable	Not Applicable	None
Settling Sampling of ISO 5 BSC E001938	Daily	01Oct26	ELB	Date of Testing (Scan)	No Growth	Not Applicable	Not Applicable	None
Settling Sampling of ISO 5 BSC E001937	Daily	01Oct26	KSM	Date of Testing (Scan C/O)	No Growth	Not Applicable	Not Applicable	None
Weekly Active Air Sampling of Cleanroom 145, 144, 143, 142 (L-Suite) with Processing BSCs								
Active Air Sampling of Cleanrooms	Weekly	01Oct26	SMO	Week of Testing	1 CFU (ISO 7 144 Sec II)	ETX-261012-0430	*Staphylococcus epidermidis*	None
					5 CFUs (ISO 8 142)	ETX-261012-0433	*Staphylococcus epidermidis* *Pseudomonas parafulva* *Bacillus safensis* *Corynebacterium xerosis* *Corynebacterium ureicelerivorans*	
Weekly Surface Sampling of Cleanroom 145, 144, 143, 142 (L-Suite) with Processing BSCs								
Surface Sampling of Cleanrooms	Weekly	01Oct26	SMO	Week of Testing	No Growth	Not Applicable	Not Applicable	None